Ordering Numbers

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Ordering Numbers

How do I put numbers in order (including decimals and negatives)?

- Use **place value** in the number to help
  - Write the numbers underneath each other, lining up their place-value columns
  - e.g. For the numbers 1 453 427 and 454 316

1	4	5	3	4	2	7
	4	5	4	3	1	6

- **Start from the highest place value (furthest left) to compare the numbers**
- For **decimals,** numbers further to the right of the decimal point are worth **less**
  - e.g. **14 is more than 8** but **0.14 is less than 0.8**
  - It can help to write the two numbers with the same number of decimal places to compare them
    - e.g. 12.115 and 12.15 are easier to compare when written as 12.115 and 12.150
- For **negative** numbers, larger values are smaller numbers
  - e.g. **14 is more than 8** but **−14 is less than −8**
  - If there is a mixture, first order the positive numbers and negative numbers separately
- **Ascending** order means in **increasing** order
  - Start with the smallest (most negative) number
- **Descending** order means in **decreasing** order
  - Start with the largest (most positive) number

Help me understand

Examiner Tips and Tricks

- Comparing numbers is easier if you rewrite them with the same number of place-value columns and in neat rows
  - e.g. To compare 213.3 and 12.245 rewrite as

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Worked Example

Write these numbers in order, with the smallest first

0.7, -0.7, 0.2991, -0.2991, 1.05, -1.05, 1.508, -1.508, 0.58, -0.58. 2.4, -2.4

Answer:

Starting with the positive numbers only
Rewrite them with the same number of place-value columns and underneath each other

0.7000

0.2991

1.0500

1.5080

0.5800

2.4000

Looking at the highest place value, there are three values starting with 0 (the smallest digit)
These are 0.7000, 0.2991 and 0.5800
Comparing their second digits gives

0.2991, 0.5800, 0.7000

Similarly, there are two values with a 1 in the highest place value column
These are 1.0500 and 1.5080
Comparing their second digits gives

1.0500, 1.5080

There is only one number with the highest place value of 2, so the list of positive numbers can now be put into order
Start with the smallest and write them as they originally appeared (without extra zeros)

0.2991, 0.58, 0.7, 1.05, 1.508, 2.4

For the negative numbers

-0.7000

-0.2991

-1.0500

-1.5080

-0.5800

-2.4000

Remembering that -2 is smaller than -1 etc
Repeat the method above to give

negative)

-2.4, -1.508, -1.05, -0.7, -0.58, -0.2991, 0.2991, 0.58, 0.7, 1.05, 1.508, 2.4

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